

Erdős Problem #562

EP-562

L1 - TRACTABLE

OPEN

Number Theory

ORIGINAL PROBLEM STATEMENT

Let $R_r(n)$ denote the r -uniform hypergraph Ramsey number: the minimal m such that if we 2-colour all edges of the complete r -uniform hypergraph on m vertices then there must be some monochromatic copy of the complete r -uniform hypergraph on n vertices.

Prove that, for $r \geq 3$, $\log_{r-1} R_r(n) \asymp_r n$, where $\log_{r-1}$ denotes the $(r-1)$ -fold iterated logarithm. That is, does $R_r(n)$ grow like $2^{2^{\dots n}}$ where the tower of exponentials has height $r-1$?

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Archive: <https://www.unsignedmath.com/problems?difficulty=1>

Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>